

Alex Lee

Waterloo, Canada | a55lee@uwaterloo.ca | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Waterloo

Sep. 2023 - Apr. 2028

Bachelor of Mathematics, Major in Data Science

Waterloo, ON

Minors: Combinatorics and Optimization, Management

Relevant Coursework: Data Structures, Algorithms, OOP, Statistics, Linear Models

SKILLS

Languages: Python, SQL(MySQL, PostgreSQL, MongoDB), R, Java, C/C++, Bash

Libraries: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, OpenCV, Matplotlib

Tools: Power BI, Tableau, Seaborn, Git, Linux, AWS

EXPERIENCE

Data Scientist

Jan. 2026 – Aug. 2026

City of Edmonton - Future Cities Institute

Edmonton, AB

- Developed **CityVision** analytics engine for **65% efficiency** gain.
- Conducted **EDA** to improve model accuracy by **15%**.
- Built **Shiny R** dashboard for real-time traffic insights.

Database Engineer

May. 2025 - Aug. 2025

Association of Korean Canadian Scientists and Engineers

Waterloo, ON

- Designed and implemented **MongoDB** schemas and **Express** APIs for the official AKCSE platform, structuring member, event, and resource data for efficient querying and future analytics.
- Optimized query patterns and indexing strategy, improving database throughput and reducing response times by approximately **3x** for core user flows.
- Implemented authentication, role-based access control, and input validation across CRUD operations to safeguard member data and maintain data integrity.

PROJECTS

FBREF Football Data Analysis | [GitHub](#)

- Collected **100,000+** football match data using **Selenium** and organized the data into a **Pandas DataFrame**.
- Identified **progress passing distance** as key predictor of match results, suggesting focused strategies to teams.

Credit Card Fraud Detection | [GitHub](#)

- Developed credit card fraud classification model using **sklearn** and **TensorFlow** with **96%** prediction accuracy.
- Processed and cleaned **200,000+** transaction records, focusing on handling imbalanced financial datasets.

Computer Vision Football Analytics | [GitHub](#) | [Link](#)

- Built a football analysis system using **OpenCV** to track key metrics such as ball possession and player speed.
- Trained a **YOLOv11** model with **10,000+** images, achieving **0.877** mAP50 with 35.7 ms per-frame latency.
- Deployed services on **AWS EC2** with **SQS**, and built a **Next.js** web app for video upload and real-time analysis.

CERTIFICATIONS

- **Machine Learning Specialization**, Stanford University | [Credential](#)
- **Deep Learning Specialization**, DeepLearning.AI | [Credential](#)